

Secure Remote Support with NetBird

Why We Recommend Controlled Network Access Instead of Unattended Desktop Access

Remote support requires a careful balance: our engineers need enough access to maintain your systems, while your organization must retain strong control over what can be reached, by whom, and for what purpose.

For ongoing managed support, we use NetBird as a secure, policy-controlled connection between authorized support systems and approved customer services. This provides a narrower and more auditable access model than permanently enabled unattended access through tools such as AnyDesk or TeamViewer.

Executive Summary

With unattended remote-desktop access, possession of the remote-access credentials can provide control over an entire desktop. Depending on the configured permissions, this may include:

- Viewing everything displayed on the screen
- Controlling the keyboard and mouse
- Running applications and administrative tools
- Accessing files and clipboard contents
- Using the customer machine to reach other systems on its local network

Our NetBird configuration follows a least-privilege model instead. Access is limited to explicitly approved systems, protocols, and ports. For example, an engineer who only needs SSH access to a Linux server can be permitted to reach TCP port 22 without receiving general desktop access or access to unrelated services.

Unattended remote desktop

Support engineer -----> Complete customer desktop
Applications, files, clipboard,
interactive user session and local network

NetBird least-privilege access

Authorized support device == encrypted tunnel ==> Approved customer service
TCP/22 only, for example

Security Benefits

1. Only Approved Services Are Reachable

NetBird policies can restrict access to the exact service required for support:

Support activity	Example permitted access
Linux administration	TCP/22 for SSH
Windows administration	TCP/3389 for RDP
Application support	Only the application's specific TCP port
Web administration	TCP/443 or the approved application port

All other ports remain unavailable through the NetBird policy unless they are explicitly approved. These ports are not opened to the public internet; they are reachable only through the authenticated, encrypted private network.

By comparison, unattended remote-desktop access commonly grants broad control of the operating system after login, even when the original support task concerns only one service.

2. Every Engineer Uses an Individual Identity

Access is tied to named support identities rather than a remote-desktop ID and password shared among multiple engineers.

- Engineers authenticate through our central identity provider.
- Multi-factor authentication can be enforced centrally.
- Access can be removed immediately when a role changes or an employee leaves.
- Customer access does not depend on distributing a reusable desktop password.

This reduces the risk of password sharing and makes administrative changes easier to attribute to an individual identity.

3. Customer-to-Customer Access Is Denied

Customer systems are placed in a destination-only group. Our access policy is unidirectional:

```
Support systems ---> Customer systems
Customer systems -X-> Support systems
Customer A       -X-> Customer B
Customer B       -X-> Customer A
```

Customer devices cannot initiate connections to support devices or to other customer devices through NetBird. Return packets belonging to a connection initiated by an authorized support system are allowed, as required for normal network communication, but this does not permit a customer system to initiate a separate connection.

4. Customer Resources Are Narrowly Defined

Where access to a machine's normal LAN address is required, we can define only that individual address as a `/32` resource. This is narrower than exposing an entire customer subnet.

Approved: 192.168.152.10/32

One specific host

Avoided: 192.168.152.0/24

Every address in the subnet

Each resource is assigned to an access policy and is unavailable to peers that are not included in that policy.

5. Traffic Is Encrypted

NetBird uses WireGuard-based encrypted tunnels between peers. When a direct peer-to-peer connection is not possible and a relay is required, application traffic remains encrypted through the relay.

The customer does not need to expose SSH, RDP, or application ports directly to the internet. The customer device establishes its own outbound connection to the NetBird infrastructure, which avoids traditional public port forwarding to the supported service.

6. Access Can Be Centrally Revoked

Access can be withdrawn without visiting every customer device to change a shared unattended-access password. Administrators can centrally:

- Remove an engineer or support device from the support group
- Disable or delete a customer peer
- Disable an access policy
- Remove a particular resource or permitted port
- Revoke enrollment credentials

This makes offboarding and incident response faster and more consistent.

7. The Control Plane Is Self-Hosted

We operate the NetBird management service within our own controlled infrastructure. We control peer enrollment, identity integration, policies, configuration backups, and management access instead of relying solely on a third-party remote-desktop control plane.

NetBird management records administrative events such as peer enrollment and changes to groups, policies, setup keys, and system settings. Host and service logs can provide additional operational evidence where required.

Comparison with Unattended Remote Desktop

Security consideration	Controlled NetBird access	Unattended AnyDesk or TeamViewer
Access scope	Explicit hosts, protocols and ports	Commonly the complete interactive desktop
Authentication	Individual organizational identity	Often a device ID and shared unattended password

Security consideration	Controlled NetBird access	Unattended AnyDesk or TeamViewer
Multi-factor authentication	Centrally enforceable through the identity provider	Product- and configuration-dependent
Customer approval per session	Not required for managed support	Not required in unattended mode
Public service exposure	Supported ports remain private behind the encrypted overlay	Remote-desktop agent connects to vendor infrastructure
Customer-to-customer communication	Denied by one-way policy	Not directly connected, but full desktop control may allow LAN access
Revocation	Central user, device, group and policy controls	Password, account, ACL or client configuration changes may be required
Least privilege	Can allow one service port only	Remote session commonly has broad operating-system control
Control plane	Self-hosted by the support provider	Commonly operated by the remote-desktop vendor

AnyDesk and TeamViewer provide legitimate security capabilities, including encryption, access control lists, multi-factor authentication and permission profiles. The concern is not that these products are inherently insecure. The important difference is the access model: permanently enabled unattended desktop control is normally broader than granting access only to a required network service.

Operational Safeguards

We combine NetBird with the following controls:

- One-way access from support systems to customer systems
- No default all-to-all policy
- No customer-to-customer policy
- Only required protocols and ports
- Individual engineer identities and multi-factor authentication
- Restricted administrator privileges
- Controlled and expiring enrollment keys
- Prompt removal of retired devices and former personnel
- Regular updates of the NetBird server and clients
- Hardened support workstations with disk encryption and endpoint protection
- Review of peer, policy, group and enrollment-key changes

What NetBird Does Not Do

NetBird is a secure connectivity layer, not a substitute for normal endpoint security. The customer system must still be patched and protected, and the service reached through NetBird must use appropriate operating-system or application authentication.

For example, permission to reach TCP port 22 does not itself authenticate an SSH user. The engineer must also possess an approved SSH identity or operating system credential. This provides an additional security boundary beyond the NetBird network policy.

Customer Control and Offboarding

At the end of a support agreement, access can be terminated by removing the customer peer from NetBird. The customer may also stop or uninstall the NetBird client locally. Once removed, the device no longer participates in the support network and the previously assigned private route is unavailable.

Conclusion

Unattended desktop tools are useful for temporary troubleshooting and can be retained as a controlled emergency option. For routine managed support, however, NetBird gives us a more precise security boundary:

An authenticated support system receives access only to an approved service on an approved customer system, through an encrypted tunnel, under a centrally managed one-way policy.

This least-privilege design reduces credential sharing, limits the accessible attack surface, prevents customer-to-customer communication, and provides a clearer basis for access review and revocation than permanently enabled access to an entire desktop.

Further Information

- [NetBird: Understanding groups and access policies](#)
- [NetBird: How routing peers and resource policies work](#)
- [NetBird: Audit events](#)
- [NetBird: Open-source security policy and advisories](#)
- [WireGuard protocol and cryptography](#)
- [AnyDesk: Interactive access, unattended access and permission settings](#)
- [TeamViewer: Security statement and unattended-access controls](#)